

TECHNICAL EVALUATION REPORT

BioCatch Connect Enterprise

Behavioral Intelligence Platform

Banking · Financial Services · Insurance (BFSI)

Confidential — Senior Technical Assessment

90+

Patents Secured

30+

Global Banks (Top 100)

16.1B

Monthly Sessions Analyzed

555M

Active Users Protected

1. Company Overview

BioCatch, established in 2011 by neural science researchers, machine learning experts, and cybersecurity specialists, is a pioneer in passive behavioral biometric technology for fraud and financial crime mitigation. Headquartered in the United States, the vendor delivers solutions tailored for the banking, financial services, and insurance (BFSI) sectors.



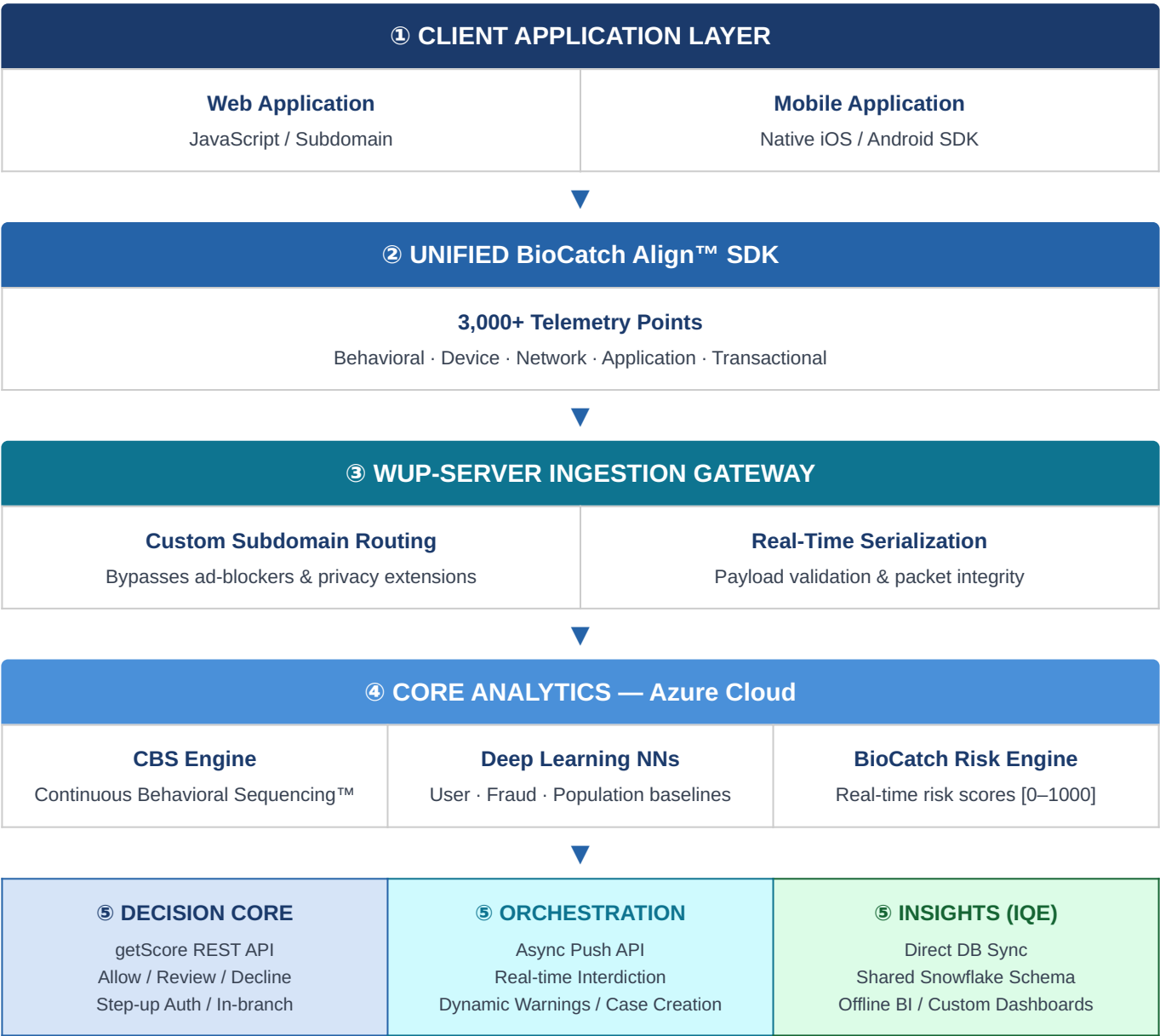
The vendor's go-to-market strategy targets 700 systemic financial institutions globally. Its FRAML (Fraud + AML) convergence platform bridges operational gaps between fraud teams and compliance departments, addressing coordinated, multi-stage attacks that siloed architectures cannot detect.

2. Product Architecture

The architectural foundation is BioCatch Connect™ 2.0, which replaces fragmented legacy SDKs with a single unified client-side ingestion module. The pipeline flows through five distinct layers from data capture to decisioning output.

Platform Architecture — Data Flow

The following diagram shows the full end-to-end data pipeline from client application to financial institution decisioning.



3. Platform Modules

The platform is organized into five specialized security modules addressing distinct threat vectors throughout the customer lifecycle.

MODULE	SECURITY OBJECTIVE	PRIMARY TELEMETRY TRIGGERS
Account Opening Protection	Prevents synthetic identity, 3rd-party theft, and automated onboarding fraud	Data familiarity · Form fluency · Navigation speed · Copy-paste actions
Account Takeover (ATO)	Detects unauthorized post-MFA access by bots, credential stuffers, remote attackers	Typing cadence shifts · Hand preference · Device angles · Remote control tools
Scams360 Protection	Detects authorized push scams where users are manipulated by external criminals	Cognitive distress · In-session pauses · Call status · Physical phone tremors
Mule Account Detection	Identifies active, sleeping, or repurposed money-laundering accounts	Login telemetry · Geographic jumps · Emulator signatures · Transaction velocity
Predictive Intelligence	Advanced analytical, visual, and relationship-mapping for fraud operations	Graphical link analysis · Session playbacks · Unified case timelines

4. Behavioral Biometrics

Unlike physical biometrics (facial scans, fingerprints) which require user action at a single point in time, behavioral biometrics run passively and continuously from login to logout, monitoring over 2,000 physical and cognitive parameters.

Physical Layer

Measures the biomechanical characteristics of user-device interaction. Captures touch area pressure, touch size, swipe trajectories, scrolling speeds, and finger-to-screen coordination. On mobile, native hardware sensors (accelerometer, gyroscope, orientation angles) construct a profile of device handling, hand steadiness, and handedness.

Cognitive Layer

Evaluates subconscious behavioral choices and application familiarity. Legitimate users show high familiarity, fluid typing, minimal corrections, and rapid navigation. Fraudsters, bots, and coached victims exhibit distinct cognitive markers:

- **Systematic field backspacing** and editing of personal details
- **High navigational hesitation** and long pauses between form fields
- **Scripted navigation** and reliance on keyboard shortcuts or clipboard copy-paste

Invisible Challenges™

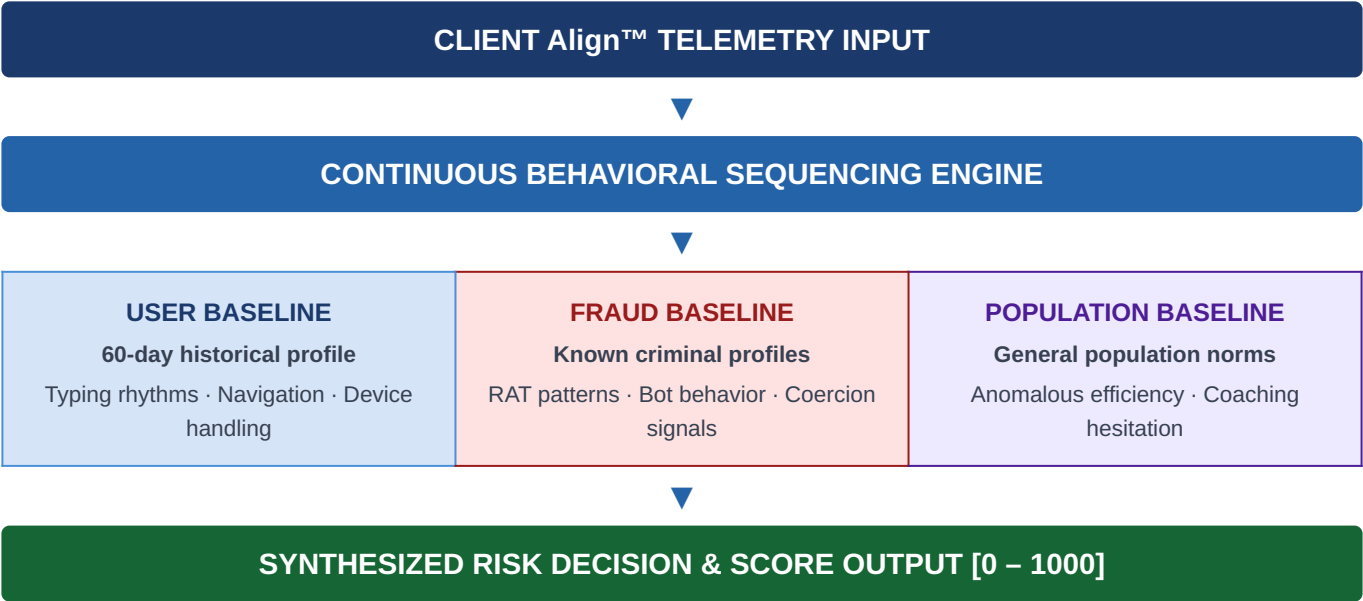
Patented passive background tests introduced in real time — such as slightly shifting a mouse cursor or introducing a millisecond delay in screen response. Human users naturally correct their coordination; automated systems, RATs, and bots either ignore the challenge or react with mechanical, non-human response profiles. Prevents replay attacks and automated credential testing.

5. Risk Engine & Scoring Methodology



Risk Engine — Continuous Behavioral Sequencing (CBS)

The CBS engine processes incoming telemetry against three distinct baseline models simultaneously, then synthesizes a unified risk decision.

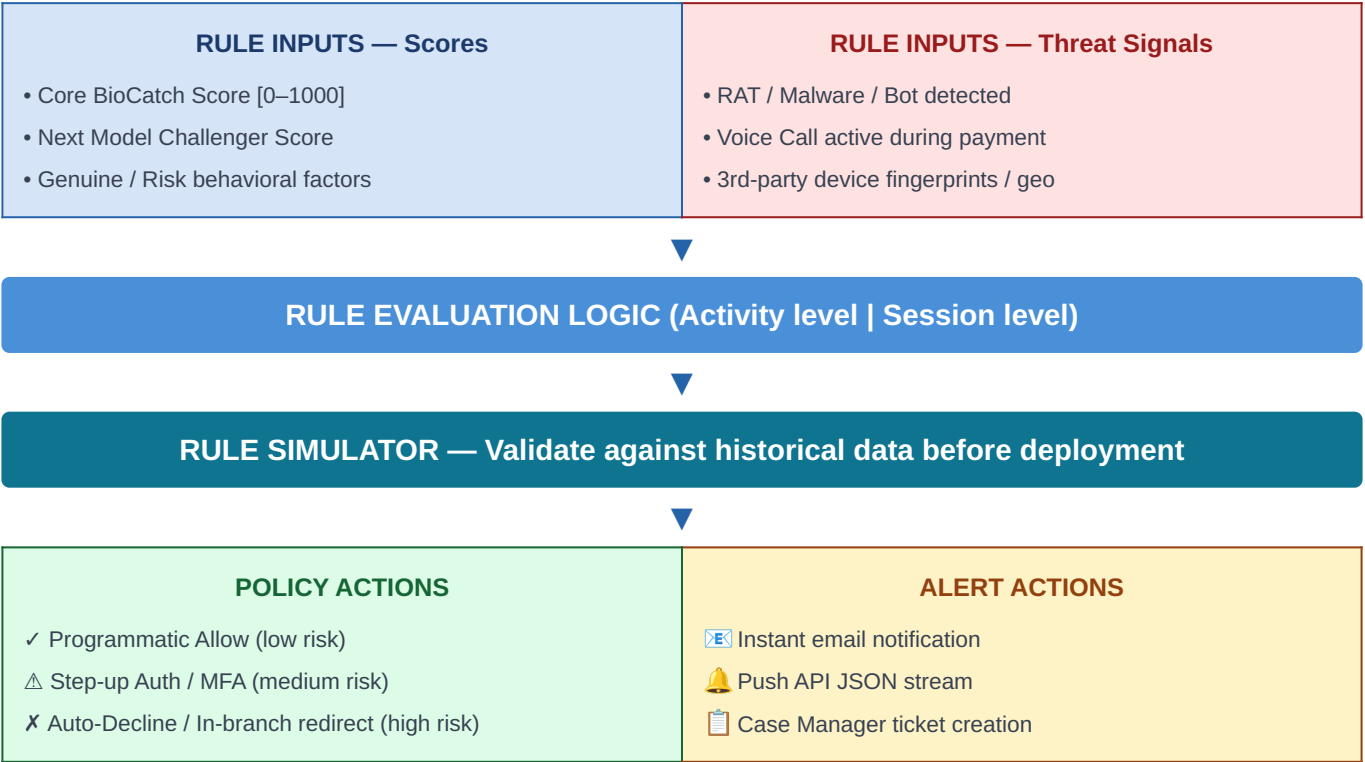


6. Rule Manager

A real-time policy configuration tool that allows risk teams to build, simulate, and deploy rules based on scoring outputs. Supports role-based access control ensuring segregation of duties between rule creation, testing, and production deployment.

Rule Manager — Policy Configuration Flow

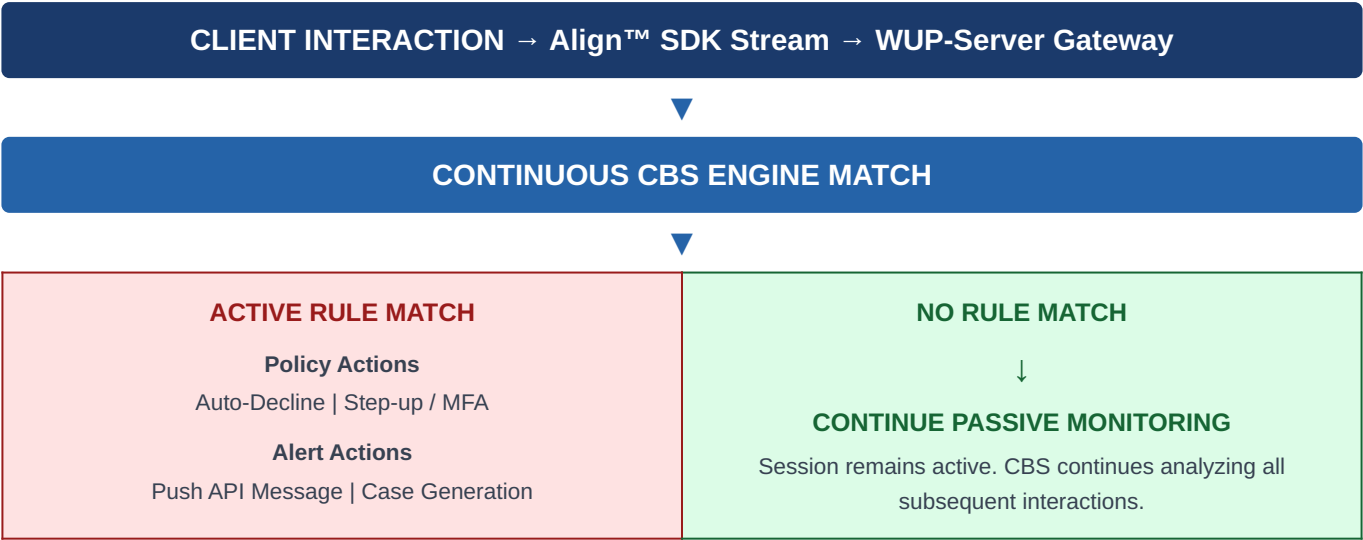
The Rule Manager converts risk scores and threat indicators into configurable policy actions and alert workflows.



7. Alert Generation Process

Engineered for sub-second response times, converting continuous telemetry streams into actionable policy outcomes.

Alert Generation Flow

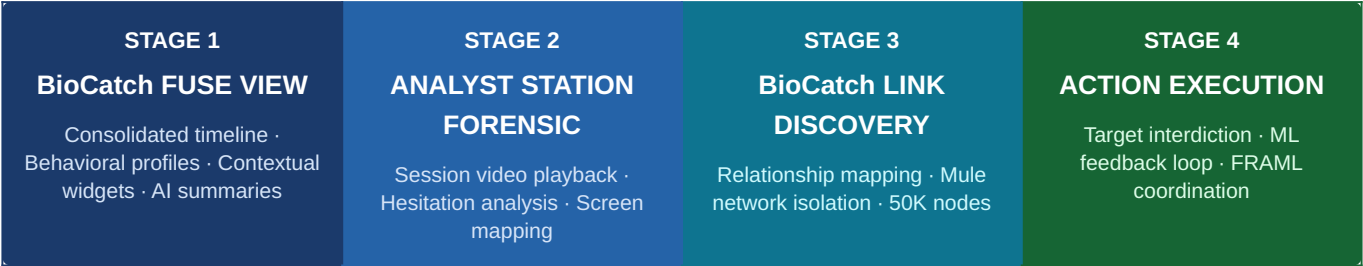


8. Analyst Station & Investigation Workflow

The Analyst Station is the primary triage and forensic portal for fraud operations teams. It provides session video reconstructions, custom query building, and contextual screen mapping to streamline incident investigation.

Investigation Workflow — Analyst Station

A four-stage structured investigation pipeline replaces reactive, transaction-by-transaction checks.



9. Device Intelligence

The device intelligence module integrates with behavioral modeling to establish physical device fingerprinting, verify session continuity, and identify technical spoofing. The Align SDK collects browser configs, OS builds, IP addresses, ASNs, and geolocation details.

- Virtual emulators cannot replicate micro-movements, device angle changes, and hand-eye coordination from physical sensors
- Monitors environment-level risks: active VPNs, proxies, and virtual machines used to hide location
- Tracks system clipboard interactions — pasted credentials vs. naturally typed indicate automated testing

10. Threat Detection Modules

10.1 Account Takeover (ATO) Detection

Continuous post-login behavioral tracking detects compromised access bypassing traditional MFA. Evaluates against the user's 60-day historical profile. Key anomalies detected:

- Sudden changes in finger touch size, pressure, and swiping speed
- Dominant hand shifts suggesting a physical imposter
- Navigational deviations from established page-sequence patterns
- Credential stuffing: automated inputs, high-speed field entries, clipboard pasting
- Post-login profile changes: password resets + payee additions + active RAT

10.2 Mule Account Detection

Targets the cash-out stage of money laundering. BioCatch maps mule accounts across five behavioral personas:

MULE PERSONA	COMPLICITY PROFILE	CORE BEHAVIORAL MARKERS
Willing Accomplice	Voluntarily participates for financial reward	Sudden velocity shifts · High data familiarity · Altered transaction patterns
Account Peddler	Sells bank accounts to criminal organizations	Sudden device/IP shifts · New behavioral biometrics on existing accounts
Misled Mule	Unknowingly participates (job/ romance scams)	High genuine history then high hesitation · Cognitive pauses · Transactional anomalies
Identity Theft	Fraudulent accounts using stolen/ synthetic identities	Expert navigation · Low typing hesitation · Highly efficient form completion
Credential Stuffing	Legitimate accounts repurposed for illicit funds	High-speed automated logins · Rapid clipboard pasting · Password resets + instant transfers

10.3 Scam Detection (Scams360)

Detects authorized push payment (APP) scams where a legitimate user is manipulated into initiating a transaction. Traditional fraud engines cannot catch these as the transaction is authorized from the genuine user's trusted device.

- **In-Session Hesitation:** Prolonged typing pauses and unusual navigation flows
- **Interaction Corrections:** Irregular typing, repeated backspacing on familiar fields
- **Coached Duress Movements:** Distinct tap timings and pauses indicating coaching
- **Active Call Detection:** Programmatic detection of active voice calls during payment flows
- **Urgency Signals:** Sudden payment amount escalations or abrupt trust transfers

10.4 Malware & RAT Detection

Identifies threats without signature databases, device IDs, or IP blacklists, detecting zero-day malware, Man-in-the-Browser (MitB) injections, and remote control tools.

- **Near-Zero Touch Areas:** Virtual tap injections vs. physical finger press
- **Lack of Physical Movement:** Absolute stillness in accelerometer/gyroscope sensors
- **In-Session Input Latency:** Temporal delays characteristic of remote control network lag
- **Application Visibility:** Identifies installed remote access software (TeamViewer, AnyDesk)
- **Scripted Inputs:** Keystroke injections and programmatic speed modifications

10.5 Bot Detection

Isolates automated scripts, hybrid software bots, and emulators targeting account creation or credential validation.

- **Physical Navigation Profiling:** Non-human typing speeds, perfect mouse tracks, systematic patterns
- **Invisible Challenges™:** Passive background tests requiring human hand-eye coordination
- **Form-Entry Speed:** Automated filling bypassing normal human reading speeds
- **Agentic AI Differentiation:** Permits legitimate AI assistants; blocks hostile scripting attacks

11. API Integrations & Data Model

API Integration Architecture

Three primary integration vectors connect BioCatch's risk engine to the financial institution's decisioning systems.

① getScore API SYNCHRONOUS REST Called at key decisioning moments: payee creation, payment execution. Returns risk score + threat indicators + behavioral factors.	② Push API ASYNC WEBHOOK STREAM Continuous risk updates from login to logout. Enables dynamic in-session warnings, step-up auth, and real-time transaction rule adjustments.	③ Alloy SDK / IQE ORCHESTRATION + SNOWFLAKE Links BioCatch sessions to KYC / identity validation. Direct Snowflake data shares for offline BI dashboards and custom analytics.
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Data Model — Core Schema Elements

- **Customer Session ID (CSID):** 10–100 character unique identifier. Alphanumeric + special chars only. Required for all API calls.
- **Screen Context Name:** Every page/view in a Single Page Application must have a unique descriptive context string for accurate risk mapping.
- **Field Tagging (data-bb):** HTML form elements tagged with data-bb attributes (e.g., data-bb="username") — tracks field-specific behavior without collecting PII.

12. Customer Onboarding

A five-phase structured onboarding process calibrates machine learning models and establishes baseline profiles before full production deployment.

Phase 1	SDK Instrumentation & Tagging: Developers integrate Align SDK, define screen context names, and apply data-bb attributes to input fields.
Phase 2	DNS & Gateway Setup: Configure custom subdomains pointing to WUP-servers to bypass ad-blockers and secure data transmission.
Phase 3	Integration Verification: Run integration tests to verify gateway packet transfer and correct context-mapping across application pages.
Phase 4	Model Calibration: ~60 days of production telemetry required to build unique user profiles and establish baseline models.
Phase 5	Feedback Integration: Configure feedback loops; mark confirmed fraud cases and false positives for continuous model training.

13. Enterprise Case Studies

Top 5 UK Retail Bank — Account Opening Optimization

CHALLENGE: Promotional digital checking account offering immediate auto-approval attracted fraudsters using stolen identities to open accounts, obtain credit, and establish money-laundering endpoints.

SOLUTION: Deployed Account Opening Protection. High-risk applicants redirected to in-branch passport authentication. Medium/low-risk continued through frictionless digital flow. Behavioral telemetry linked to email metadata revealed cross-session identity fraud.

RESULTS: £800K saved in first three months · Thousands of fraudulent applications flagged · Hundreds of citizens notified of identity compromise · Led to global rollout across North America, Canada, and APAC.

Top Latin American Retail Bank — Scam Reduction

CHALLENGE: Surge in social engineering scams where customers were coached over phone calls to make authorized push transfers.

SOLUTION: Implemented BioCatch mobile behavioral protection focusing on duress indicators, swipe hesitation, and call status monitoring.

RESULTS: 67% reduction in total social engineering fraud losses without increasing friction for legitimate customers.

Large Global Financial Institution — Operational Efficiencies

CHALLENGE: Required an ML-driven fraud mitigation engine capable of adapting to credential compromise and remote access attacks across all digital channels.

SOLUTION: Implemented continuous session monitoring across all mobile and web banking channels.

RESULTS: 98% fraud detection rate · ~\$2 million in monthly savings across fraud losses and operational expenses.

14. Competitor Market Comparison

Five-vendor evaluation across primary technology focus, telemetry depth, analysis context, target audience, and pricing model.

Vector	BioCatch Connect 2.0	LexisNexis ThreatMetrix	SEON	Sift	Arkose Labs
Primary Focus	Passive behavioral biometrics & cognitive analysis	Shared consortium identity databases	Digital footprinting & social media lookup	AI-driven transactional risk scoring	Interactive challenge-response (CAPTCHA)
Telemetry Depth	3,000+ continuous interactions per session	78B+ identity records + basic device signals	Multi-source API (IP/email/phone/ 300+ social)	Global transaction data + basic device behavior	Challenge metadata + session markers
Analysis Context	Continuous session-long passive profiling	Point-in-time checks at login/ transaction	Point-in-time API lookup	Real-time transaction scoring	Dynamic feedback during login loops
Target Audience	Enterprise banks & high-risk FIs	Enterprise corps with dedicated fraud teams	Industry-agnostic (SMEs to Enterprise)	Digital merchants, e-commerce, fintechs	Large web properties fighting automated bots
Pricing	Premium; 60-day calibration required	Premium; complex database integration	Moderate; fast onboarding (2 days)	Scale-based; API onboarding	Usage-based; active endpoint integration

15. Conclusions & Recommendations

Strategic Assessment

BioCatch crystalline Connect 2.0 provides an enterprise-grade solution for financial institutions transitioning from reactive, point-in-time fraud checks to continuous behavioral-based session protection. By leveraging Continuous Behavioral Sequencing and cognitive biometrics, the platform detects sophisticated threat vectors — authorized push payment scams, RATs, and automated bots — that bypass traditional MFA and password-based controls.

The premium pricing, UI dependencies, and 60-day calibration period mean the platform is best suited for mid-to-large-scale financial institutions with dedicated fraud analysts and structured onboarding capabilities.

Actionable Recommendations

1. **Unify Ingestion with Align™** — Migrate to Connect 2.0 SDK to simplify maintenance, reduce latency, and capture enriched telemetry.
2. **Implement Subdomain Routing** — Configure custom subdomain routing during onboarding to bypass ad-blockers and secure session data.
3. **Configure Next Model Score Challenger Loops** — Run challenger models in parallel with production rules to evaluate performance without operational disruption.
4. **Deploy Push API for Real-Time Warnings** — Integrate the async Push API to trigger dynamic in-app warnings that help break fraudster psychological hold during scam coercion.
5. **Bridge Fraud and AML (FRAML) Teams** — Leverage BioCatch Link visual mapping to break silos between fraud and AML departments, accelerating mule investigations.